



PHYSICAL & ELECTROMAGNETIC SECURITY FOR AI DATA CENTRE BUILDS

CYBERSHIELD

**Protect the AI data centre.
Contain the signal.**

A measurable electromagnetic security boundary —
engineered, integrated and verified as one system.



HOW WE THINK ABOUT SHIELDING

For decades, the building has had to serve the shield.

THE STATUS QUO

For as long as shielded rooms have existed, the facility has had to accommodate the shield. Conventional shielding is welded into place, fixed at design stage and permanent from the day it is finished. The building is planned around it. Construction stops for it. And when racks change, power density rises or the site outgrows itself, the room cannot follow. Protection ends up being the least flexible part of the fastest-moving infrastructure there is.

Engineering heritage since 1987

5 global locations

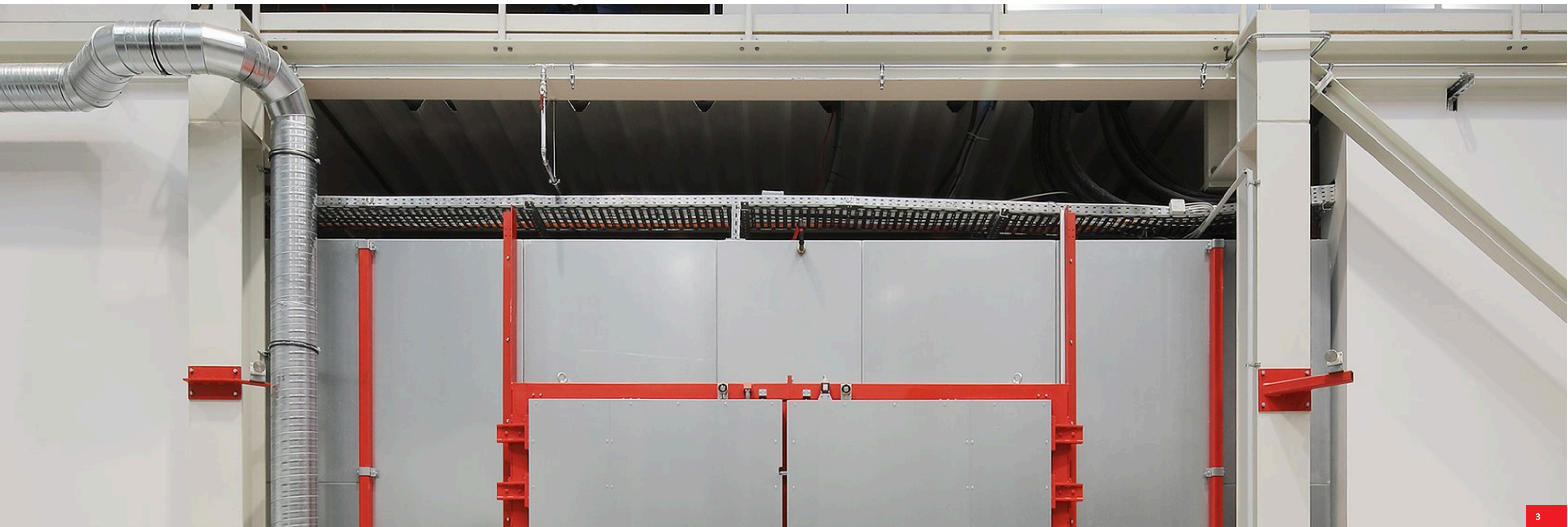
Presence in 80+ countries

Turnkey delivery

WHAT WE BELIEVE

We believe it should be the other way around: the room adapts to the facility, not the facility to the room. So we build ours from prefabricated steel modules that pass through a standard building door, assemble from the inside, sit close to existing walls and bolt together — no welding, no glue, nothing irreversible. The room can be extended, reconfigured or relocated entirely, then measured again.

Full protection, none of the permanence.



AI data centre security no longer ends at the software layer.

Firewalls, encryption and zero trust stop what arrives over the network. As the value held inside an AI data centre grows, the paths that never touch the network — through physical space and through electromagnetic coupling — have become a real part of the assessment. Software security and physical security now have to be designed together.

0 dB

The attenuation firewalls, encryption and zero trust provide against a signal that never enters the network.

01 A national strategic asset

AI compute is already treated as national capability. Model weights, training data and sovereign workloads are corporate property and a matter of state interest at once — which is exactly what makes them worth targeting.

02 EMC and EMP exposed by design

Dense GPU racks take tens of kilowatts through switching electronics, and 400G/800G interconnects work to noise budgets measured in millivolts. As power density rises, emission goes up and immunity headroom comes down. Protection sized for a conventional server room does not cover this.

03 Protection from what is outside

Data centres sit near industry, transmitters and transport infrastructure. The ambient RF environment is not yours to control and only gets busier. Intentional interference can be assembled from commercially available parts — which is why IEC 61000-4-36 exists as a test standard in its own right.

HOW THE EXPOSURE ARRIVES

Four routes to the same asset — none of them across a firewall.

Each one reaches processing hardware through physical space or through electromagnetic coupling. Each one is closed at the facility boundary.

Compromising emanations

Sensitive processing activity can be exposed through unintended electromagnetic signals — without touching the network.

POTENTIAL IMPACT

Confidentiality exposed

Intentional interference

Localized high-power RF or electromagnetic energy can disrupt electronics, controls and communications.

POTENTIAL IMPACT

Service interruption

EMP / HEMP exposure

Radiated and conducted pulse effects can challenge critical systems and continuity architectures. The E1 pulse rises in 2.5 ns — before surge protection rated for lightning has reacted.

POTENTIAL IMPACT

Mission continuity risk

Boundary vulnerabilities

Doors, ventilation, power, data, cooling and utility penetrations can become the weakest path through the shield.

POTENTIAL IMPACT

Protection degraded

WHO CYBERSHIELD IS BUILT FOR

Three environments where the boundary has to hold.

Hyperscale cloud & AI centres

Core AI compute clusters, quantum hardware and critical availability zones, shielded against RF tampering and high-power electromagnetic threats.

Colocation & enterprise data centres

A certified shielded vault offered as a premium, measurable security tier for enterprise customers under strict governance requirements.

Defence, government & financial institutions

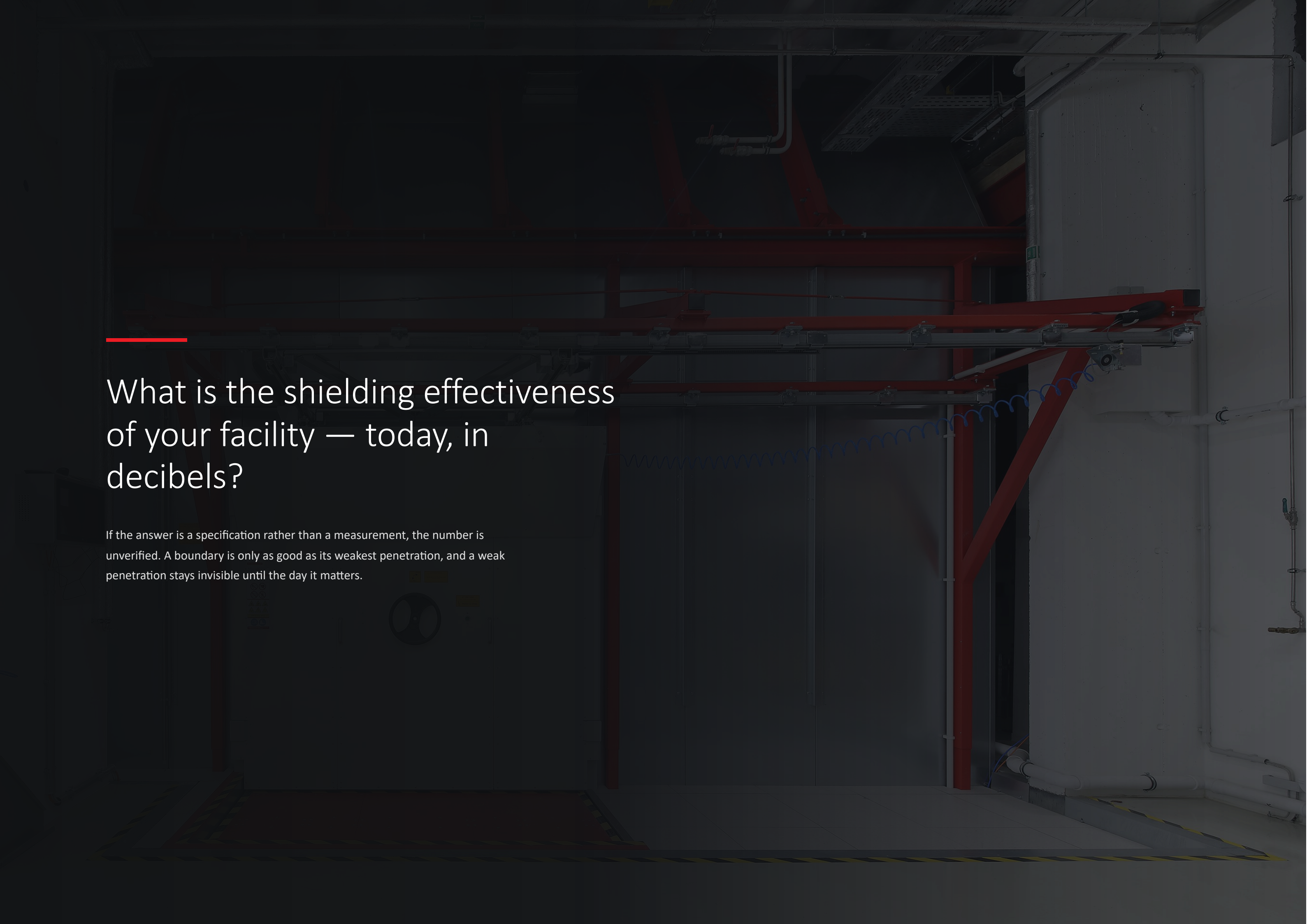
Sovereign cloud deployments, command centres and high-frequency trading platforms where confidentiality cannot be left to assumption.

WHAT THE BOUNDARY HAS TO COVER

- › Core compute halls and high-density accelerated racks
- › Key management, crypto and administration rooms
- › Network entry points, meet-me rooms and cross-connects
- › Power rooms, UPS lines and every filtered service crossing



Physical and electromagnetic exposure begins at the site boundary, not at the rack.



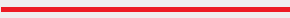
What is the shielding effectiveness of your facility — today, in decibels?

If the answer is a specification rather than a measurement, the number is unverified. A boundary is only as good as its weakest penetration, and a weak penetration stays invisible until the day it matters.

THE SHORT ANSWER

One boundary — designed, built and measured under a single responsibility.

Shielding structure, doors, filters, ventilation waveguides and every penetration are engineered as one continuous boundary, assembled weld-free alongside live operations, then measured on site to EN 50147-1 / IEEE 299 and handed over as documented evidence. What you keep at the end is a measurement, not an assurance.



MODULAR PAN SHIELDING SYSTEM

A secure room engineered around the reality of your facility.

The room is built from prefabricated PAN modules — bolted, reversible and engineered around the building it goes into rather than the other way around.

01 Precision assembly

Panels are bolted every 75 mm with predefined torque and conductive mesh gaskets.

02 Architectural integration

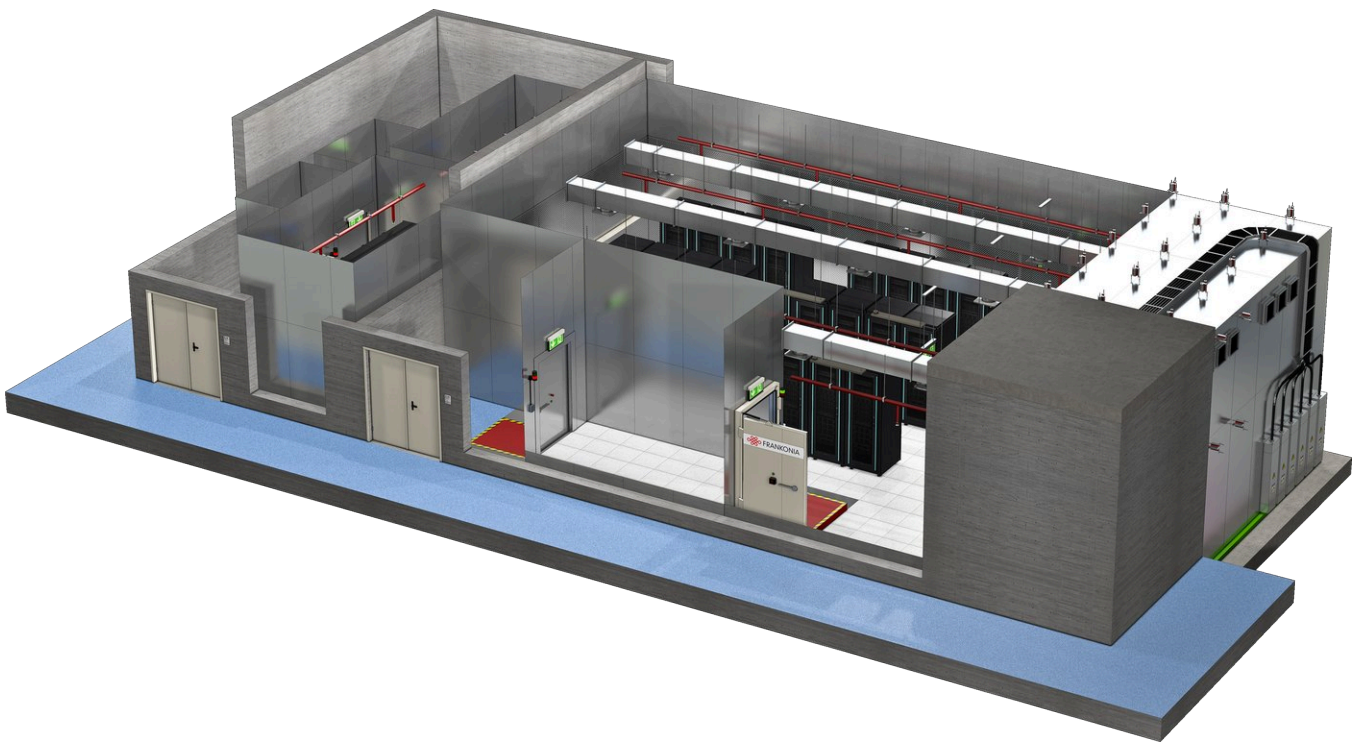
Reversible modules leave flat interior surfaces for finishing walls, ceilings and racks, designed around raised floors, fire systems, lighting, cooling and access control.

03 Adaptable by design

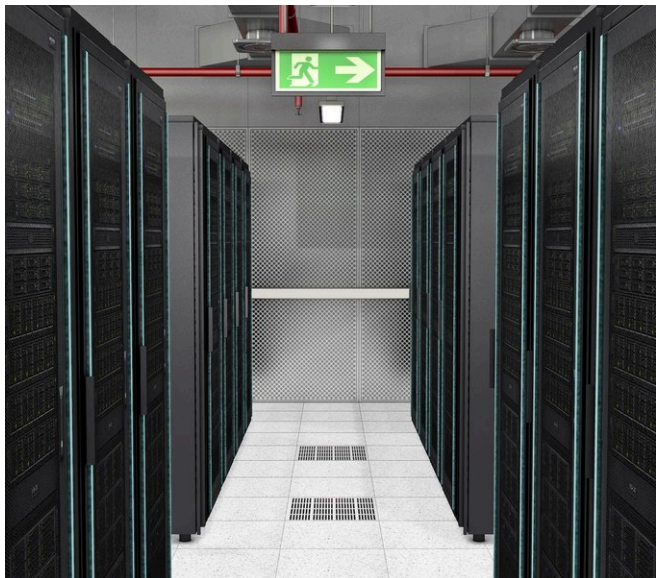
Dismountable without damage for expansion, modification or complete relocation.

04 Complete boundary

Shielding structure, doors, filters, honeycombs and waveguides are treated as one system.



Shielded data hall with power room, sluit, filtered penetrations and raised floor — engineered as one envelope.



Cold-aisle containment inside the shielded volume.



Personnel sluit with HF doors and access control.



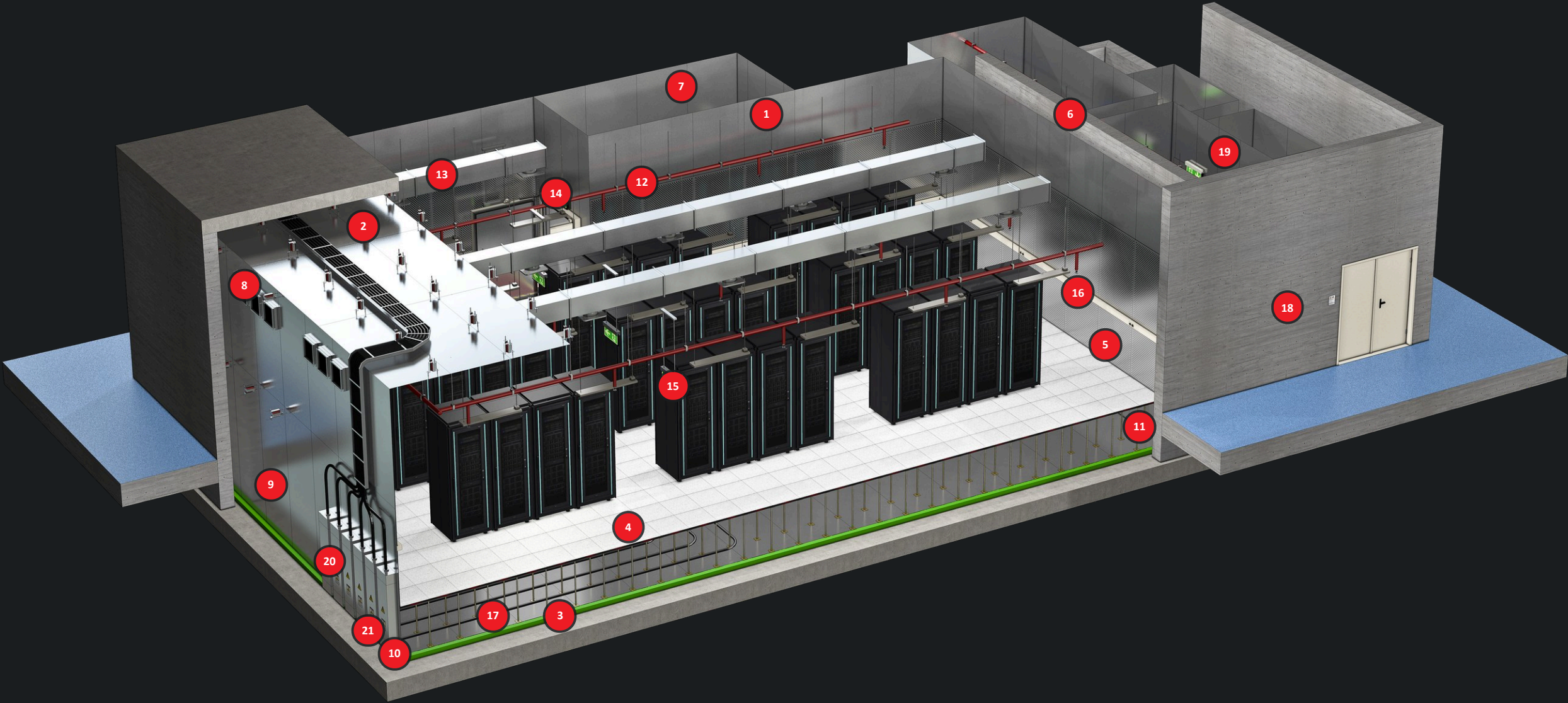
PAN TYPE SHIELDING, IN BRIEF

- › Shielding modules made of 2.0 mm thick galvanized steel
- › Self-supporting stability, or a static steel structure for any seismic condition
- › Mounted from the inside; reverse installation gives a flat interior surface
- › Raised floor systems or welded floor systems
- › Dismountable without any damage — easy modification and maintenance

EXPLORE THE ROOM

See what CyberShield builds into a shielded room.

Twenty-one engineered solutions make up the shielding boundary — structure, penetrations, building services and power. Each one is designed, supplied and measured as part of the same envelope.



STRUCTURE

- 1 Modular prefabricated PAN construction
- 2 Isolated wall and ceiling fixation
- 3 GFRP isolation grid
- 4 Modular raised floor to 5 t
- 5 Integrated acoustic panels
- 6 Fire prevention, room-in-room

PENETRATIONS

- 7 Sluice system with HF doors
- 8 Ventilation honeycombs
- 9 Custom access hatches
- 10 Shielded media interfaces to DN200
- 11 EMP ducting between rooms
- 12 Fire-prevention water and gas feeds
- 13 AC fluid feeds and ducting

BUILDING SERVICES

- 14 Access control and signalling
- 15 Emission-free LED lighting
- 16 Internal electrical installation
- 17 Integrated cable and media ducts
- 18 EMP-hardened motor units

POWER AND EARTHING

- 19 Self-sustaining power room
- 20 AC/DC power, EMP and HEMP filters
- 21 Advanced earthing

VERIFIED, NOT ASSUMED

Performance you don't have to take on faith.

Every CyberShield project ends with measured evidence. Shielding effectiveness is tested on site according to international standards, and documented acceptance results are handed over with the room. Detailed performance data is available for your engineering team.

MEASURED AND VALIDATED AGAINST

EN 50147-1 Shielding effectiveness measurement	IEEE 299 Available as a project option
BSI TL-03305 / 03306 Eavesdropping-protected rooms & IT enclosures	NATO SDIP-27 Level A TEMPEST design & approval, aligned with NSA 94-106
MIL-STD-188-125-1 / -2 HEMP and IEMI protection, project-specific validation	ISO/IEC 27001 Supports the physical and environmental security controls



Shielding effectiveness is measured after installation and handed over as documented acceptance evidence.

EVIDENCE BASE

Figures, terminology and threat definitions in this brochure are read from the publications below. Each entry names the issuing body, the document and what it supports.

IEC 61000-2-9 Electromagnetic compatibility — Part 2-9: Description of HEMP environment, radiated disturbance Cited for: the E1 HEMP waveform and the 2.5 ns rise time quoted in this brochure
IEC 61000-4-36 Electromagnetic compatibility — Part 4-36: IEMI immunity test methods for equipment and systems Cited for: intentional electromagnetic interference treated as its own test discipline
NCSC TEMPEST and electromagnetic security guidance Cited for: the definition and scope of compromising emanations
NIST SP 800-53 Security and Privacy Controls for Information Systems and Organizations — control PE-19, Information Leakage Cited for: information leakage as a physical and environmental protection control
CISA Resilient Power Best Practices for Critical Facilities and Sites Cited for: power resilience practice in critical facilities
EN 50600 Information technology — Data centre facilities and infrastructures Cited for: data centre facility and infrastructure terminology

WHAT YOU RECEIVE AT HANDOVER

- › On-site shielding effectiveness measurement across the specified frequency range
- › Leak detection sweep of doors, filters, honeycombs and every feed-through
- › Documented acceptance report issued with the room
- › Periodic re-testing and security re-certification as a service

≥ 90 dB at 10 kHz, magnetic field	≥ 120 dB at 100–400 MHz, plane wave	≥ 100 dB at 40 GHz, microwave
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Where standard shielding stops, the engineering starts.

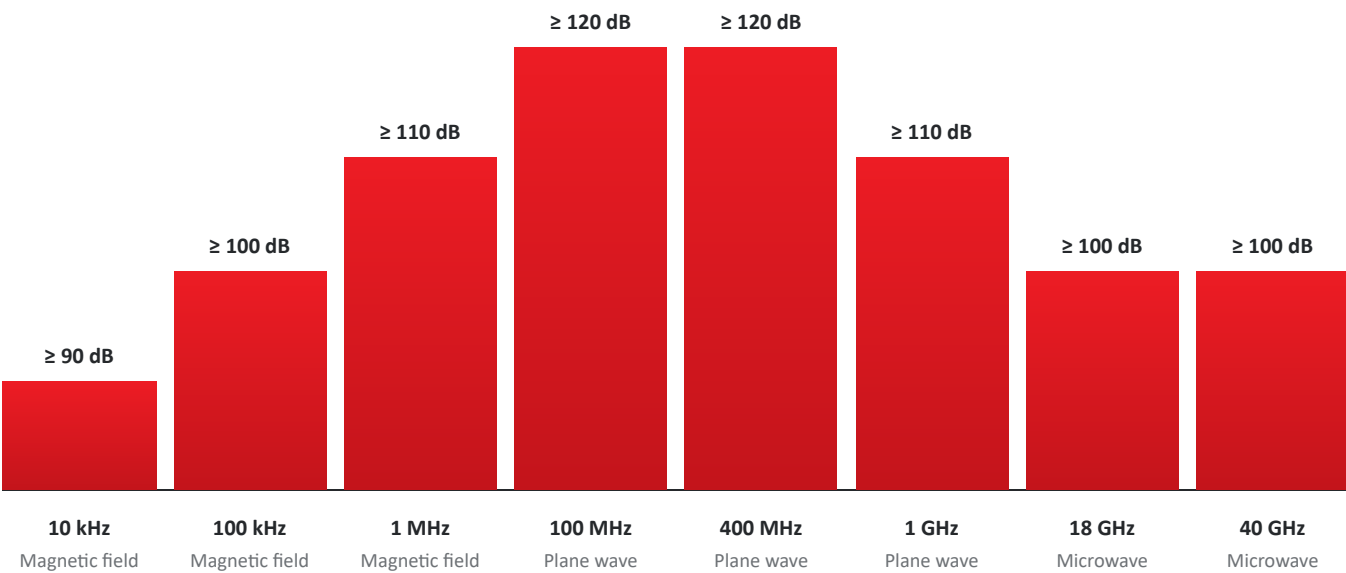
How CyberShield compares with conventional shielding approaches across the criteria that decide long-term performance.

PERFORMANCE CRITERION	CONVENTIONAL SHIELDING	FRANKONIA CYBERSHIELD
RF attenuation spectrum	60–80 dB across a narrow frequency band	Peak ≥ 120 dB (100–400 MHz) and ≥ 100 dB from 10 kHz to 40 GHz
Shielding panel engineering	Thin sheet metal or welded steel construction	2.0 mm galvanized steel PAN modules bolted every 75 mm to defined torque
Joinery & sealing technology	Conductive adhesive or continuous welding	High-conductivity mesh gasket — no glue, no welding, 100 % reusable
Airflow & acoustic efficiency	Basic cut-outs with thermal loss and RF leakage paths	Honeycomb waveguide ventilation and ISO 354 sound absorption ($\alpha = 0.65$)
Building integration	High static loads within a permanent structure	Self-supporting or seismic steel structure, dismantlable without damage



One shielding envelope, verified from 10 kHz to 40 GHz.

Standard-setting PAN type module engineering, measured in accordance with EN 50147-1 and IEEE 299. The same attenuation is engineered into every door, filter, honeycomb vent and feed-through in the boundary.



Values describe the guaranteed performance envelope of the standard PAN type system. The scope that applies to your project is confirmed in the specification and by on-site acceptance testing.



Every penetration is filtered to the level the wall itself achieves — the envelope is only as good as its weakest crossing.



Customized

ONE CONTINUOUS BARRIER

Six product lines. One zero-leak boundary.

Every component is engineered as part of the same shielding envelope, so performance is not lost at the joints, the doors or the penetrations.



01

CyberShield Structure

Prefabricated 2.0 mm galvanized steel PAN module system for walls, ceilings and floors.

Sheet steel to DIN 17162 / EN 10142 DX 52 D+Z, 275 g/m² galvanising, bolted every 75 mm onto a self-supporting steel structure.



03

CyberShield Connectivity

High-performance power line filters, fibre-optic waveguide penetrations and RF signal suppressors.

Power, data and signal feed-throughs filtered to the same attenuation as the wall they cross — up to DN200 media penetrations.



02

CyberShield Access

Heavy-duty sliding and hinged RF doors, high-attenuation RF windows and integrated access monitoring.

Circumferential triple contact system with conductive springs, 20,000 opening cycles MTBF, 150 mm standard threshold.

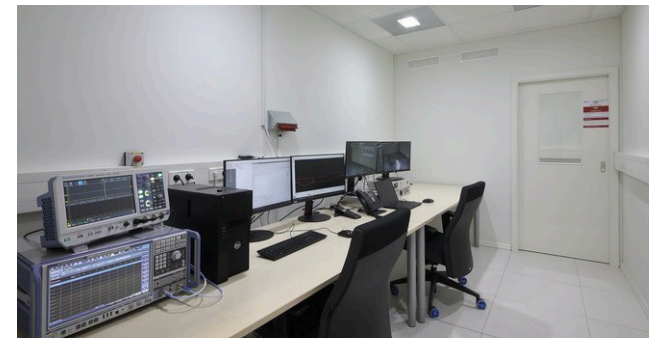


04

CyberShield Air & Waveguides

Honeycomb ventilation panels, acoustic panels (ISO 354) and shielded waveguides for liquid cooling and utilities.

Below-cut-off honeycomb waveguides pass air without passing RF; acoustic inlay panels absorb to ISO 354, $\alpha_w = 0.65$ (MH).



05

CyberShield Validation

EN 50147-1 / IEEE 299 shielding measurement, leak detection, SE testing and compliance documentation.

Guaranteed shielding effectiveness of ≥ 90 dB at 10 kHz rising to ≥ 120 dB at 100–400 MHz and holding ≥ 100 dB to 40 GHz.



06

CyberShield Lifecycle

Preventive maintenance, recalibration and periodic re-certification services.

Nothing glued or welded, so the room can be dismantled without damage — modified, extended or relocated, then re-measured.

COMPLETE SCOPE, ONE SUPPLIER

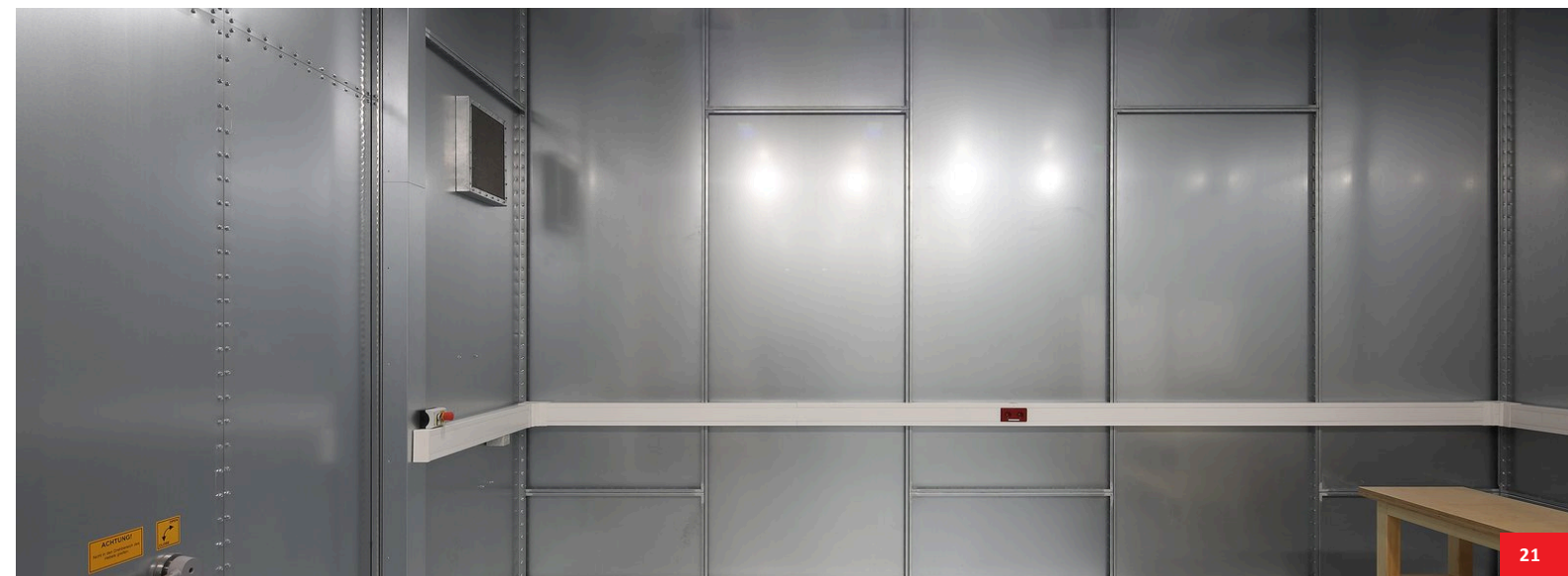
As a specialist in RF shielding, Frankonia offers a broad range of complementary, standardized and customized products to maintain its position as a turnkey provider. Every room is custom designed with its own full-integrative electrical system setup and simple interfaces to the facility.

ELECTRICAL INTEGRATION

- › Distribution unit, cabling, safety functions per local standards
- › LED lighting, explosion protection option, emergency lighting
- › AC and DC filters, signal and data filters, optic converters
- › Safety matrix and higher-level control unit (PLC)
- › CE conformity per Machinery Directive 2006/42/EG

VENTILATION, SMOKE AND GAS

- › Honeycombs, cooling and exhaust systems
- › Air sampling network for gas and smoke detection
- › Detection analyzer with alarm central; ATEX compliance
- › Liquid detection system
- › Extinguishing solutions (sprinklers, etc.)



HOW THE ROOM IS CONFIGURED

One platform.
Four mission profiles.

01

GOVERNMENT & SOVEREIGN CLOUD

Sovereign Compute Vault

Create a controlled processing zone for classified or nationally sensitive workloads.

02

AI LABS & HYPERSCALE OPERATORS

AI & HPC Shielded Zone

Protect high-value models, training data and accelerated compute infrastructure.

03

COLOCATION PROVIDERS

Colocation Shielded Vault

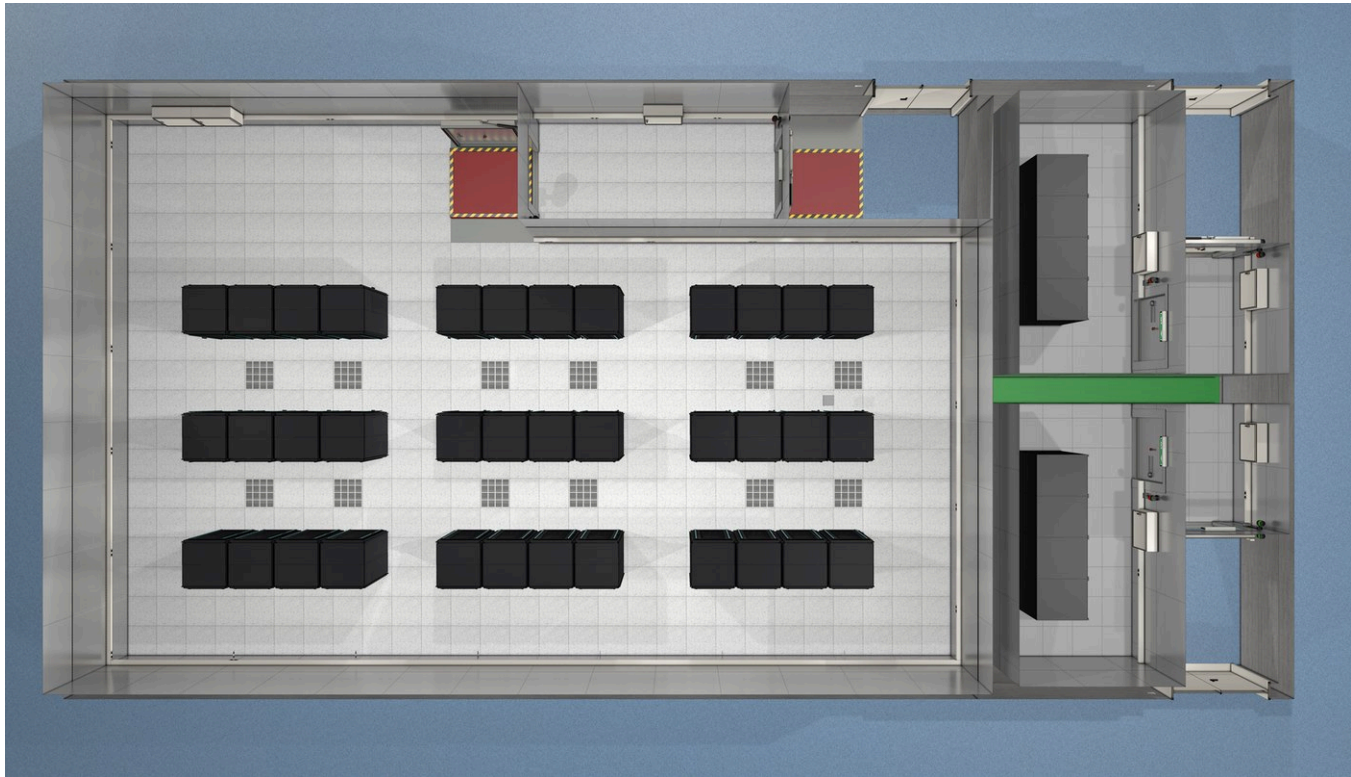
Offer a measurable premium security tier for regulated enterprise customers.

04

DEFENCE, FINANCE & COMMUNICATIONS

Mission Continuity Suite

Support continuity architectures exposed to electromagnetic disruption risks.



Plan view of a shielded data hall: rack rows, containment, filtered power wall and adjoining power room.

HOW CYBERSHIELD IS USED

Three situations we
see again and again.

GOVERNMENT & SOVEREIGN CLOUD

A classified zone inside a live facility

CHALLENGE	A government cloud programme required a secure processing zone for classified workloads inside an existing data centre.
APPROACH	The shielded vault was engineered around live operations, installed without welding and verified on site.
OUTCOME	A high-security zone delivered and acceptance-tested without interrupting the surrounding facility.

AI LAB

A measurable boundary around model assets

CHALLENGE	An AI company needed to protect model weights and training clusters against emanation and interference risks.
APPROACH	A dedicated shielded hall integrated cooling, power and monitoring for high-density racks.
OUTCOME	A measured, documented security boundary around the company's most valuable IP.

COLOCATION

A premium security tier without rebuilding

CHALLENGE	A colocation operator wanted a premium tier for regulated customers — without rebuilding the site.
APPROACH	A modular vault was added inside existing white space and packaged as a verified product.
OUTCOME	A new revenue stream from a security tier competitors cannot easily match.

Representative scenarios based on typical project profiles.



WHAT HAPPENS WHEN YOU REACH OUT

From first conversation to verified protection.

- 01

Initial consultation
A specialist reviews your goals, site and constraints. No documentation is required to start.
- 02

Risk & site assessment
Assets, threat scenarios, ambient RF conditions and facility constraints are assessed; protection requirements are defined together.
- 03

Concept & 3D engineering
You receive a concept design, CAD/BIM architectural integration and a transparent quotation.
- 04

Precision manufacturing
PAN modules, RF doors, power and data filters and honeycomb vents are produced in Frankonia's own facilities.
- 05

Installation & verification
Certified teams assemble the modules without welding; shielding effectiveness and leak detection are measured and documented at handover.
- 06

Operation & support
Preventive maintenance, periodic re-testing and security re-certification keep the boundary effective over time.

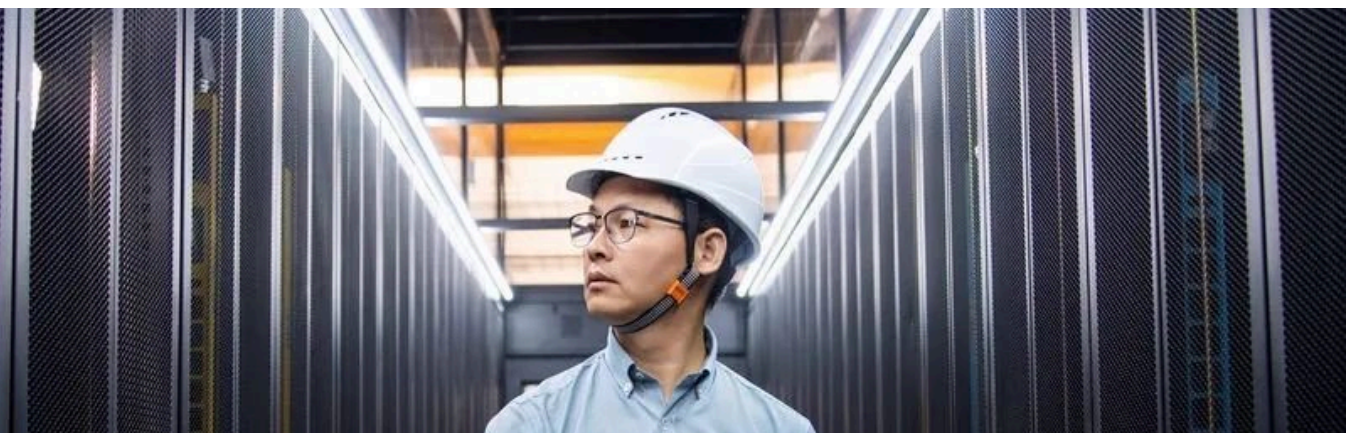


CLEAR SCOPE FROM DAY ONE

What Frankonia delivers, and where partners take over.

A transparent split of responsibility across the protected room, so nothing inside the boundary is left to assumption.

CATEGORY	IMPORTANCE	SCOPE & EXPERTISE
HF shielding performance	CORE	In-house manufacturing with guaranteed performance
Doors & access solutions	CORE	Precision RF hinged and sliding doors
Filtered penetrations	CORE	Power, data and signal RF filters
Ventilation waveguides	CORE	Precision honeycomb airflow systems
EMC testing & certification	CORE	On-site validation and leak detection
Maintenance & service	SERVICE	Annual audits and lifecycle support
Building security systems	PARTNER	Interface integration with facility security
Intrusion alarm systems	PARTNER	Interface integration with room alarm systems
Organizational procedures	CUSTOMER	Best-practice advisory provided by Frankonia



In-house project management, engineering, production, installation and service — the boundary and its acceptance measurement come from one source.



Reliable

WHAT MAKES THE DIFFERENCE

Advantages & benefits.

Frankonia is a top-notch solution provider for EMC solutions from Germany, with an all-encompassing range of products to customize every project with perfection and on time.

PRODUCT ADVANTAGES

Modular and self-supporting

The shielding carries itself, so the parent building takes no additional structural load.

Identical performance throughout

The same shielding effectiveness across panels, doors, filters and waveguides — no weak link at the joints.

Bolted, never glued or welded

Every joint is a screwed connection with a conductive mesh gasket, so the room stays serviceable for decades.

Produced in-house

Prefabricated standard components manufactured by Frankonia rather than sourced and assembled.

CUSTOMER BENEFITS

Proven for more than 35 years

Technologies in service in EMC test laboratories worldwide since 1987.

Few demands on the building

The design adapts to existing electricity, statics and building conditions rather than the other way round.

Relocate, modify, re-sell

The modular setup can be dismantled and rebuilt elsewhere without damage.

Simple building interfaces

Electricity, ventilation, exhaust, data and gas detection cross the boundary through defined, filtered interfaces.

WHY CUSTOMERS SELECT US

One partner, start to finish

Consultation, manufacturing, installation, acceptance testing and recertification under a single responsibility.

Building design support

Frankonia works with architects and contractors early, while shielding decisions are still inexpensive to make.

No compromise on safety

Non-combustible, recyclable materials — no carbon, no polyethylene, no glue.

Project management that finishes

Experienced teams carry both the detail and the schedule through to handover.

COMMON QUESTIONS

The questions every project starts with.

Can CyberShield be installed in an existing, operating facility?

Yes. Modules pass through standard doors and are bolted together from the inside — no welding or hot work — so installation alongside live operations is a common project profile.

How long does a project take?

It depends on size and integration scope. As a rough guide, room assembly itself takes weeks, and the full cycle from assessment to verified handover typically runs several months. A concrete schedule is part of the concept proposal.

What determines the cost?

Size, performance requirements, the number of penetrations (doors, power, data, cooling) and integration complexity. Basic project parameters are enough for a first consultation and estimate.

Does the shield interfere with cooling, fire safety or daily operations?

No. Airflow, raised floors, fire suppression, lighting and access control are engineered into the boundary from the start.

What if we need to expand or relocate later?

The system is dismantlable without damage. Rooms can be extended, reconfigured or reassembled at a new site — protecting the original investment.



START WITH YOUR RISK PROFILE

Let's define the right protection boundary.

Tell us what you need to protect. A Frankonia specialist will review your project and respond directly. Basic project parameters are enough to begin — no documentation is required to start the conversation.

sales@frankoniagroup.com
www.frankonia-cybershield.com

FRANKONIA AT A GLANCE

The shielding is not new. The application is.

The Frankonia Group was founded in 1987 as a solution provider for EMC and antenna test laboratories, and is today a specialised technology corporation for anechoic chambers and test systems in the automotive, military and industrial sectors. CyberShield applies that same shielding engineering to data infrastructure. In-house project management, engineering and production — with Frankonia's own installation and service teams — mean the boundary, the doors, the filters and the acceptance measurement all come from one source.



Frankonia Group headquarters and production site, Heideck, Germany.

FRANKONIA STANDS FOR

Global presence

A well-structured network of production, representation and service units, active worldwide.

Innovation as method

Technologies adopted to raise efficiency, outcome and quality alongside customer needs.

Complete solution provider

Fundamental knowledge across every discipline a shielded facility touches.

Preferred partner

Customised, state-of-the-art solutions rather than catalogue products.

CORE VALUES

Ingenuity

We never stop pushing forward. We look for advantage in everything, handle challenges with creativity and strive for simplicity.

Community

We act fair and respectful, roll up our sleeves to get the job done, shape the future together and win as a team.

Energy

We bring energy to everything we do, stay hungry, embrace opportunities and share the passion for what we do.

Fairness

We are open minded and curious, admit failures and improve, communicate with clarity and honesty and take responsibility.

SALES & INFORMATION

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Verified on site

Shielding performance is measured after installation — not assumed.

Weld-free assembly

Bolted modules installed alongside live operations, with no hot work.

Built to change

Dismount, expand or relocate the room without damage.

CyberShield is a high-assurance engineering solution. Performance, standards and certification scope depend on the agreed project configuration and final validation.





FRANKONIA

Your Shielding Expert from Germany

Unique customer solutions for your security.

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